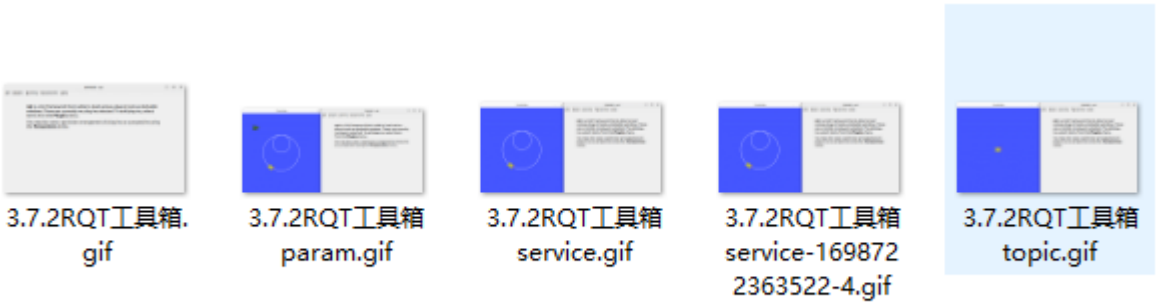
18. ROS2 Rqt Toolbox

- The folder of this section contains a demo animation in git, so you can intuitively see the operation flow of the cases in this section



1. Introduction to Rqt

Rqt is another modular visualization tool provided by ROS. Just as its name suggests, like Rviz it is also developed on top of the QT visualization toolkit. Before using it, we need to install it with the following command, and then it can be started and used with the rqt command.

2. Installation

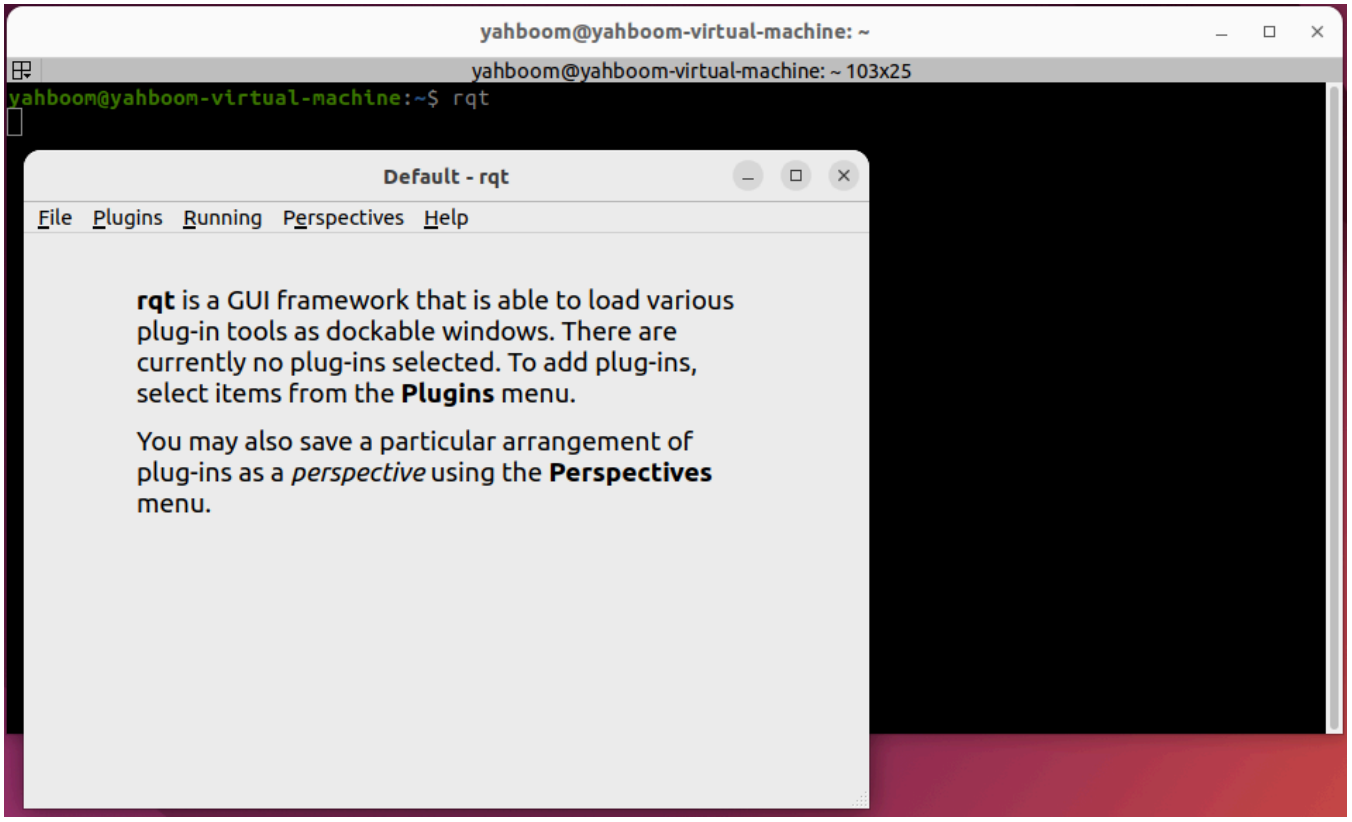
- Generally, as long as the desktop version is installed, the rqt toolbox is installed by default;
- If ROS was not installed as the full version, you can install it in the following way

```
sudo apt install ros-${ROS_DISTRO}-rqt*
```

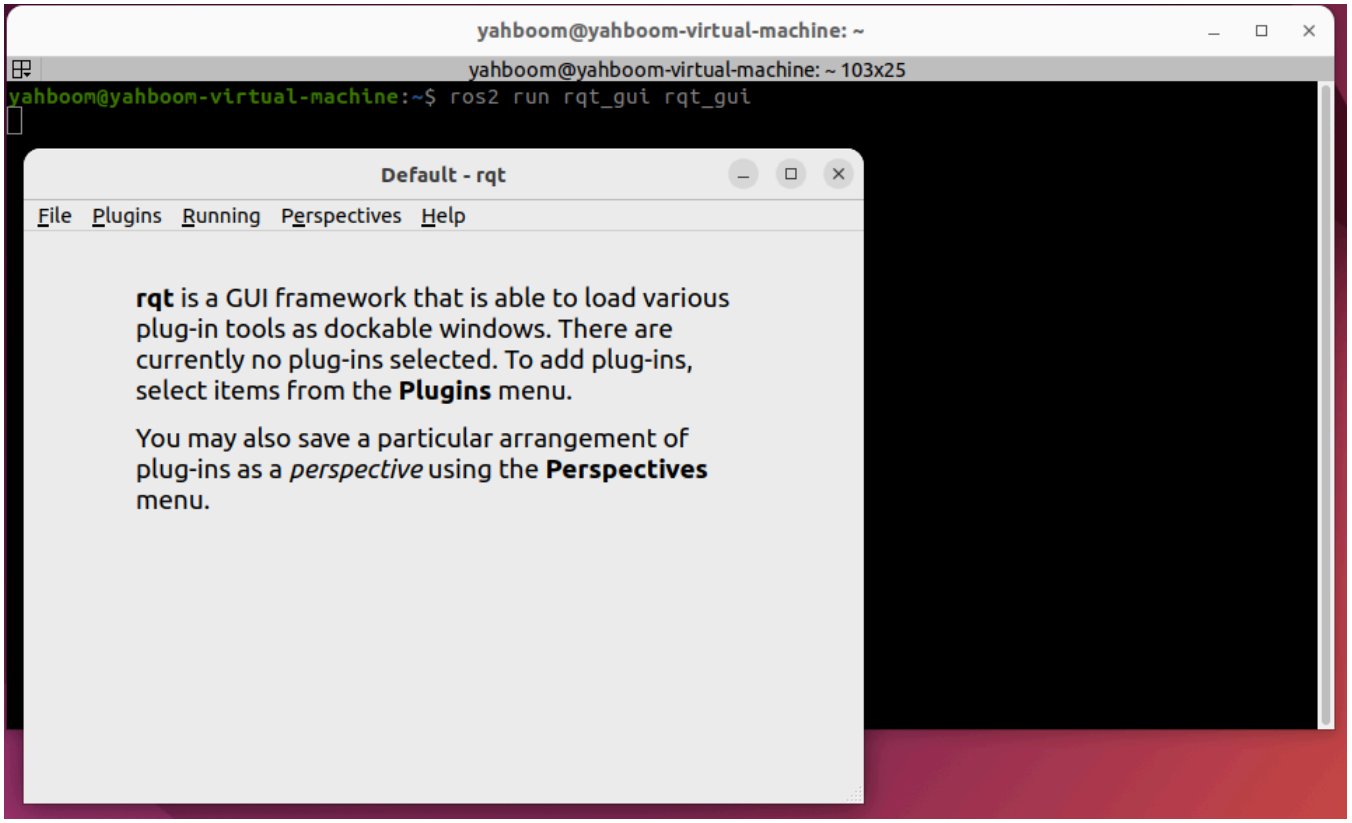
3. Startup

The common `rqt` startup commands are:

- Method 1: `rqt`

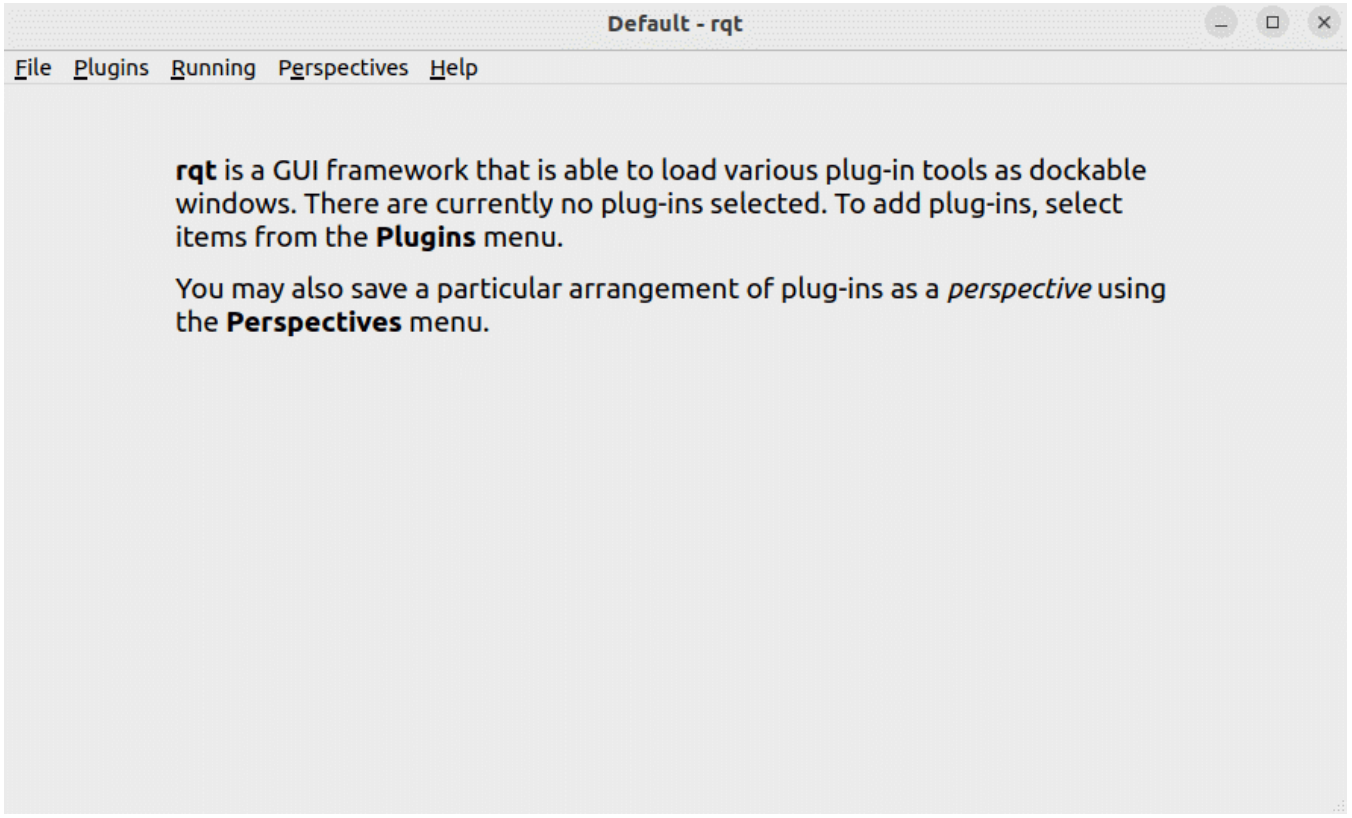


- Method 2: `ros2 run rqt_gui rqt_gui`



4. Using the Plugins

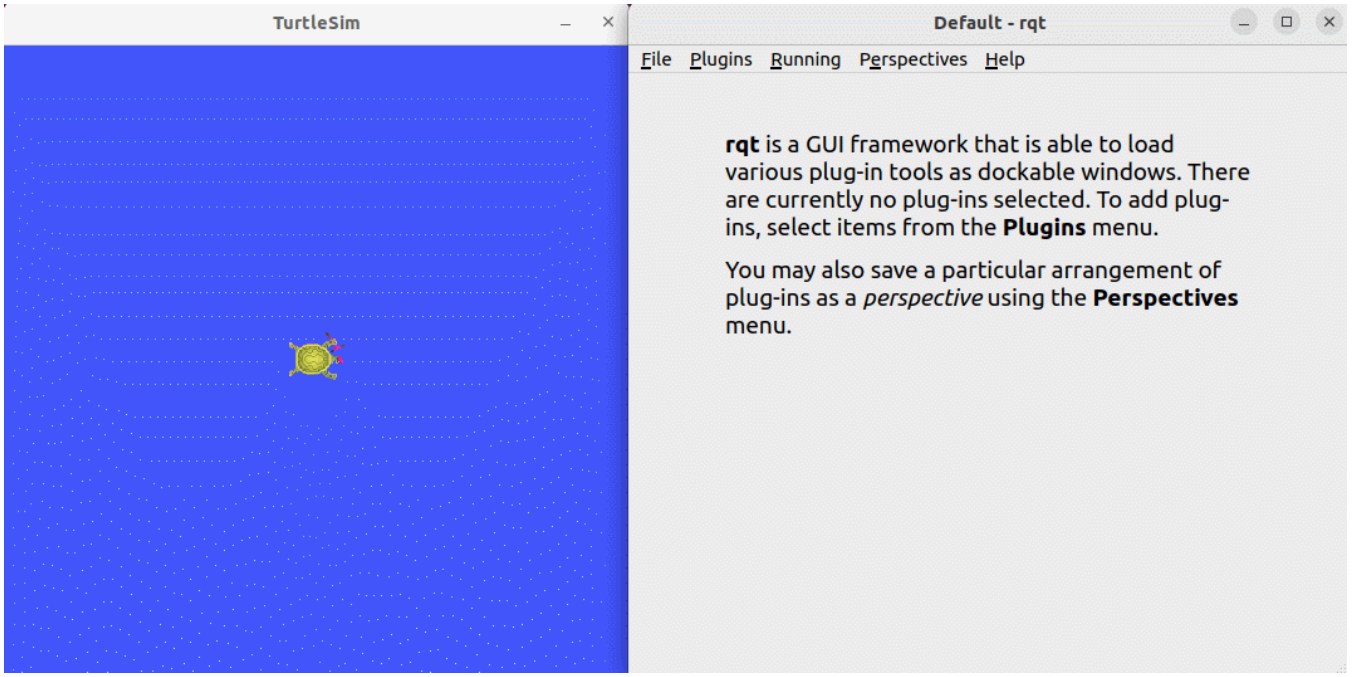
After rqt has started, you can add the plugins you need through plugins:



The plugins include the ones related to topics, services, actions, parameters, logs, and so on. You can select them as needed, which makes it convenient to debug ROS2 programs. The usage examples are as follows.

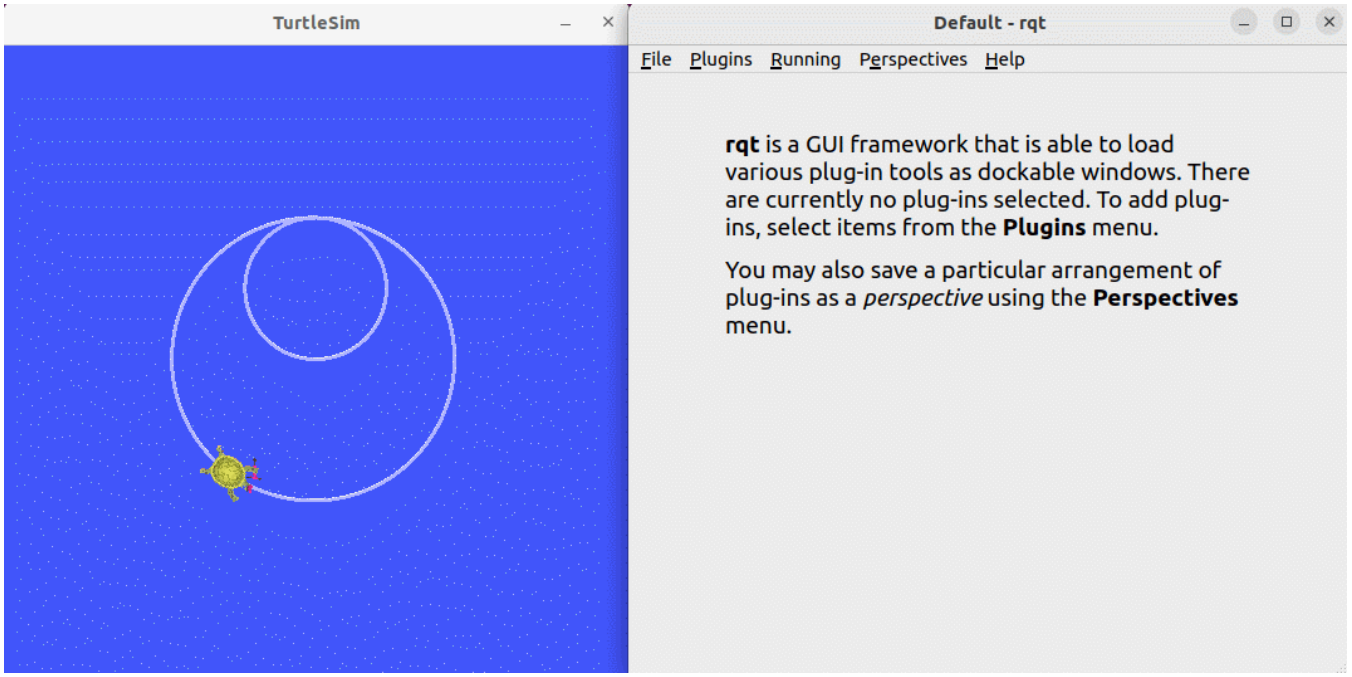
4.1. The topic plugin

Add the topic plugin and send a velocity command to control the movement of the turtle.



4.2. The service plugin

Add the service plugin and send a request to spawn a turtle at the specified position.



4.3. The parameter plugin

Dynamically modify the background color of the turtle window through the parameter plugin.

